

MYSSP · MALAYSIA SCIENCE SCHOLAR'S PROGRAMME

PROJECT P-1 · TSUNG-DAO LEE INSTITUTE, SJTU

Seeing through solid rock

Finding a hidden chamber with cosmic-ray muons

A research handbook for your muography project

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Eight weeks · 8 hours a week · remote, with a weekly meeting with your mentor

How to use this handbook

Three of you, one mentor, and eight weeks to turn a cosmic-ray shower into a picture of something nobody has X-rayed before. This handbook exists so that your first week is spent doing physics rather than guessing what your mentor meant.

What this project actually is

Every second, cosmic rays hit the upper atmosphere and shower the ground with muons — free, everywhere, and able to cross rock that stops almost everything else. Point a detector at a mountain (or a pyramid) for long enough, count how many muons survive, and the places where more of them got through are the places with less material in the way. That is muography, and it is how the ScanPyramids collaboration found a hidden void inside the Great Pyramid of Khufu in 2017. Your project is to build that measurement, end to end, as a simulation in Python: a virtual target with a hidden chamber, a beam of simulated muons arriving from many viewing directions within a finite field of view, expected counts followed by a realistic noisy dataset, and a reconstructed image that shows — or fails to show — the chamber. Nobody in this project touches a real detector. The underlying muon-transport physics is the same either way — but a real detector also introduces acceptance, efficiency, background, detector resolution, and a sky that is not perfectly steady, none of which a simulation gives you for free. Chapter 6 asks you to be honest about that gap in your own report.

Who this project suits

You do not need to already be a strong programmer. (The original project listing named strong programming skills as a preferred background — in practice, prior Python experience helps you move faster, but it is not required, provided you are comfortable learning independently.) You start from scratch in your understanding, but not from scratch in the code: week 1 teaches the Python you need, while the supplied starter code provides the physics and geometry you will extend. What it does reward is curiosity across fields — this is physics, statistics, and code, in roughly equal measure, and the students who enjoy it most tend to be the ones happy to move between those three rather than sticking to one. Using an AI assistant to help you learn the tools is completely fine (Chapter 4 has more on doing that honestly).

The three parts

Part I is the physics you need before you write a line of code: what a muon is, why it gets through rock at all, and why counting survivors reveals what is hidden. **Part II** is your project itself, worked through in the five steps you will actually code, plus the statistics you need to answer the question every mentor asks: *are you sure, or did you just get lucky?* **Part III** is the practical side — your eight-week schedule, exactly what you hand in and when, and what to do before week one. The appendices at the back collect every formula and every install command in one place, so you never have to hunt back through a chapter to find one.

The boxes

Colour carries meaning. You never need to read a box's title to know what kind of thing it is.

BOX

WHAT IT MEANS

■	Definition	A term used precisely from here on. Use it loosely in your report and you will be asked to fix it.
■	Key result	A formula or fact you will actually use in your code. Numbered, so you can cite it in your report.
■	Example	A worked number. Do the arithmetic yourself once, by hand, before you trust the code to do it.
■	Note	An aside, a transition, or the closing paragraph of a chapter.
■	Intuition	The picture to keep in your head once the algebra is gone.
■	Think about it	A question with no printed answer. Bring it to your weekly meeting.
■	A bit of history	Where the idea came from. Physics is argued into place, not handed down.
■	Watch out	A mistake that has already cost somebody else in this group real time.
■	Checklist	End of chapter. If you cannot tick every line, do not move on yet.

The dark, gold-barred block at the head of every chapter — **In a nutshell** — is the heaviest thing on the page, on purpose. It gives you the chapter in five sentences and the terms it will spend. If you read nothing else on a busy week, read those.

Notation

m_μ	muon mass, 105.66 MeV/ c^2	ρ	material density
ϱ	opacity, $\int \rho \, d\ell$	$E_{\min}$	minimum energy to survive a path
γ	relativistic factor, $E/m_\mu c^2$	θ	zenith angle, from straight overhead
$\left\langle -\frac{dE}{dx} \right\rangle$	mean energy loss per unit path	σ	Poisson (statistical) uncertainty

Note Everything in this handbook about dates, meeting times or who is doing what is provisional until your mentor confirms it. Everything about the physics is not.

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PART I —

The idea

A particle that falls on you for free, and what it can tell you

A particle that falls on you for free

In a nutshell

In a nutshell. The muon is a heavier, unstable cousin of the electron. Two facts about it matter for this project: it is *minimum-ionising*, which means it loses energy slowly and can cross metres of rock before it runs out of steam; and it arrives free and continuously — about one muon crosses every square centimetre of horizontal ground every minute, day and night, at no cost to you. Together, those two facts are the entire reason muography works.

Key terms. muon · minimum ionising particle (MIP) · stopping power · range · cosmic-ray flux · zenith angle · $\cos^2 \theta$ law.

You already know the electron. For this project, the most useful way to think about the muon is as a much heavier cousin of it — same charge, same basic behaviour in an electric or magnetic field, but different in mass, lifetime, and what that difference in mass makes possible.

1.1 A heavier charged lepton that does not last

The muon is a distinct particle, not literally an electron with a different mass — but it shares the electron's electric charge, and behaves the same way in almost every respect except mass, lifetime, and the handful of properties that follow from those two.

Definition 1.1: The muon

$$m_e = 0.511 \text{ MeV}/c^2, \quad m_\mu = 105.66 \text{ MeV}/c^2 \approx 207 m_e.$$

The muon is unstable: left alone, it decays after a mean lifetime of $\tau_\mu = 2.2 \mu\text{s}$,

$$\mu^- \longrightarrow e^- + \bar{\nu}_e + \nu_\mu.$$

Two microseconds sounds like nothing, and for a muon sitting still it would be. But cosmic muons are not sitting still.

A bit of history Carl Anderson and Seth Neddermeyer found the muon in a cloud chamber in 1936 while looking for something else entirely. Nobody had predicted a second, heavier electron, and nobody knew what it was for. The physicist I. I. Rabi is said to have responded, on hearing the news, with the single most famous sentence in the muon's history: “Who ordered that?” Ninety years later, the honest answer is still: we do not fully know — and that is exactly why physicists keep measuring it.

A fast-moving particle's clock runs slow, as seen from the ground — a consequence of special relativity called *time dilation*. How much slower depends on a number called the relativistic factor, $\gamma = E/m_\mu c^2$: the particle's total energy divided by its rest energy. A 4 GeV cosmic muon has $\gamma \approx 4000/105.66 \approx 38$, so from our point of view on the ground it lives about 38 times longer than the $2.2 \mu\text{s}$ it would live at rest — roughly $\gamma\tau_\mu \approx 83 \mu\text{s}$ — long enough to cross about 25 km of atmosphere before it decays.

Intuition You do not need the full machinery of relativity to use this: the bigger γ is (that is, the more energetic the muon is compared with its own rest mass), the longer it effectively lives as seen from the ground, and the further it gets before decaying. A muon with ten times the energy lives, and travels, roughly ten times further for exactly this reason.

That time dilation is a major reason sufficiently energetic cosmic-ray muons survive the trip to the ground at all — a muon without it would decay high in the atmosphere, long before reaching a detector.

1.2 Why it gets through rock

A charged particle crossing matter loses energy mainly by knocking electrons out of the atoms it passes. Cosmic-ray muons at GeV energies happen to sit close to the point where that rate of loss is at its lowest — the *minimum-ionising* regime.

Definition 1.2: Minimum ionising particle (MIP)

A charged particle moving through matter in the region where its rate of ionisation energy loss is near the smallest it can be. For ordinary materials, that minimum is of order

$$\left\langle -\frac{dE}{dx} \right\rangle \approx 2 \text{ MeV per g cm}^{-2},$$

an approximation rather than a fundamental constant — the exact value depends mildly on the material. This number is quoted per unit *mass* thickness (grams per square centimetre), which is a deliberately density-independent way of talking about “how much stuff.” Multiply by a material's density to get MeV lost per centimetre: about 5.3 MeV/cm in typical rock ($\rho \approx 2.65 \text{ g cm}^{-3}$), roughly 5.0 MeV/cm in limestone ($\rho \approx 2.5$, the value used for the pyramid's limestone in your code skeleton).

Key result 1.1: Range–energy relation

Ignore the (small, at these energies) losses from radiation, and a muon’s energy falls off linearly as it ploughs through material. Its range — the mass thickness it can cross before running out of energy — is approximately

$$R(E) \approx \frac{E}{\left\langle -\frac{dE}{dx} \right\rangle} \quad [\text{g cm}^{-2}].$$

Turn this around and it tells you the *minimum energy* a muon needs to survive a given opacity ϱ of material:

$$E_{\min} \approx \varrho \times \left\langle -\frac{dE}{dx} \right\rangle.$$

This is the second half of the flux-and-range calculation your project runs in week 2, and it is the physics underneath every muography image ever made.

Note This linear rule is the right level of detail for building intuition, and it is accurate enough for the numbers in this chapter. Your code skeleton does something slightly more careful — it solves for $E_{\min}$ without throwing away the radiative term, which matters more at higher opacities. The exact version, and why the two agree closely here, is in Appendix A.

Example 1.1: How far does a typical cosmic muon get?

The mean cosmic muon at sea level has an energy of about 4 GeV. Using Key result 1.1,

$$R \approx \frac{4000 \text{ MeV}}{2 \text{ MeV g}^{-1}\text{cm}^2} = 2000 \text{ g cm}^{-2} \approx 7.5 \text{ m of rock.}$$

Seven and a half metres — not a kilometre. To cross 100 m of limestone ($\rho \approx 2.5$, i.e. $2.5 \times 10^4 \text{ g cm}^{-2}$) a muon needs

$$E_{\min} \gtrsim 2 \times 2.5 \times 10^4 \text{ MeV} \approx 50 \text{ GeV},$$

more than ten times the typical energy. A mountain, or a pyramid, is imaged not by the ordinary muon but by the rare, high-energy tail of the spectrum — which is exactly why muography exposures are measured in days, weeks, or months, and never in minutes.

Watch out “Muons cross a kilometre of rock” is a sentence you will hear, and it is true only of the very rare, very high-energy muons. If you size an exposure using the *typical* 4 GeV cosmic muon, you will underestimate the running time by orders of magnitude, because almost none of those typical muons survive a real target. Your simulation has to use the full energy spectrum, not a single typical energy — that is exactly what Appendix A gives you.

1.3 The flux you get for free

Primary cosmic rays — mostly protons from outside the Solar System — slam into the upper atmosphere, make pions and kaons, and those decay into muons. What reaches sea level is a continuous, all-day and all-night rain that nobody has to switch on. It is not perfectly constant in reality — it drifts by a few per cent with air pressure, temperature, and solar activity — but for this educational simulation, we will treat it as constant. A real experiment has to measure and correct for those variations; your report’s discussion section is a good place to say so.

Key result 1.2: Cosmic muon flux at sea level

For muons of momentum above 1 GeV/c, the intensity arriving from straight overhead is

$$I_v \approx 70 \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1},$$

which is the origin of the well-known rule of thumb: roughly one muon per square centimetre of horizontal ground per minute. Away from the vertical, the intensity falls off with zenith angle θ (measured from straight overhead) as

$$I(\theta) \approx I_v \cos^2 \theta.$$

This is the “provided formula” for your first plot: flux against zenith angle, from directly overhead ($\theta = 0$) out towards the horizon.

Note That third unit, sr^{-1} (“per steradian”), is worth pausing on if you have not met it before. A steradian is the 3D version of an angle: instead of measuring how wide a flat wedge is (degrees, radians), it measures how wide a *cone of directions* is — how much of the sky a patch of view actually covers. “Muons per second, per square metre, per steradian” just means: for every little cone of incoming directions, here is the rate. You do not need to calculate in steradians yourself for this project; you only need to know that the unit is there because direction, not just area and time, is part of what the flux depends on.

Strictly, I here is an *intensity* rather than a plain “flux” (the sr^{-1} is the giveaway). This handbook follows common usage in muography and calls it “flux” throughout, the way most of the field does casually — worth knowing, for whenever you read the professional literature later, that the two words are not always interchangeable.

Intuition A muon arriving from near the horizon has to cross far more atmosphere to reach you than one arriving from overhead — more of its parent pions decay in flight or interact before ever making a muon, and more of the muons that do form decay or lose too much energy along the long slanted path. Fewer survive the trip. That is the whole reason for the $\cos^2 \theta$ shape: it is a statement about *how much sky is in the way*, not about muons themselves being reluctant to travel sideways.

Think about it The formula above holds down to $\theta = 0$ and up to roughly 70° from the vertical. What do you think happens as $\theta \rightarrow 90^\circ$ – and why might a real detector pointed at the side of a mountain need to worry about it? Bring your answer to your first weekly meeting.

Note We can now say what the particle is: heavy enough that its journey through rock is a real physics problem, penetrating enough to be a usable imaging beam, and free. What we cannot yet say is how counting survivors turns into a picture of something hidden. That is the next chapter – and it is the chapter your project is actually about.

Before you move on

- I can state m_μ , τ_μ , and roughly how far a 4 GeV muon gets through rock, from memory.
- I can explain, in one sentence, why cosmic muons are called minimum ionising.
- I understand why the mean cosmic muon cannot image a mountain, and what kind of muon can.
- I can state the flux formula $I(\theta) \approx I_v \cos^2 \theta$ and explain the shape in words.
- Glossary: **MIP**, **stopping power**, **range**, **zenith angle**.

Seeing through solid rock

In a nutshell

In a nutshell. Point a muon telescope at something and count how many muons survive the trip through it. Fewer survivors means more material in the way; more survivors means less. A hidden chamber is, to a muon, just a patch of sky with slightly less rock in front of it — and that patch sends you slightly more muons than its solid-rock neighbours. Everything in this project is that one idea, worked out carefully enough to tell a real signal from ordinary bad luck.

Key terms. opacity · absorption muography · counting statistics · signal significance · exposure time.

2.1 Opacity: how much stuff is in the way

Along any line of sight through a target, a muon has to cross a certain amount of material.

Definition 2.1: Opacity

$$\varrho = \int \rho \, d\ell \quad [\text{g cm}^{-2}],$$

the density integrated along the path. A muon needs energy $E_{\min} \approx \varrho \times \left\langle -\frac{dE}{dx} \right\rangle$ (Key result 1.1) just to reach the far side. Less opacity along a direction means a lower energy threshold, which means more of the cosmic-ray spectrum can survive that direction — which means *more muons arrive from it*, on average, than from a direction with more rock in the way.

Intuition Think of the target as a stained-glass window and the muon flux as sunlight behind it. Thick, dense glass lets less light through; thin glass, or a hole, lets more through. You are not looking at the window directly — you are standing behind it, counting how much light reaches you from each little patch, and inferring the thickness of the glass in front of each patch from how bright it is. A hidden chamber is a thin patch in an otherwise thick window.

A bit of history Luis Alvarez and collaborators placed spark chambers in the Belzoni Chamber beneath Khafre's pyramid in 1967–69, looking for hidden chambers, and — importantly — found nothing (Alvarez *et al.*, *Science* **167**, 832, 1970). The technique was vindicated almost exactly fifty years later: in 2017, the ScanPyramids collaboration reported a large void above the Grand Gallery of Khufu's pyramid, seen independently by three different detector technologies (Morishima *et al.*, *Nature* **552**, 386, 2017). A null result and a discovery, half a century apart, from the same idea. Both were good physics — the second one just had more muons.

2.2 From opacity to a predicted count

Your simulation's job, in one sentence, is to turn a density map into a predicted number of muons per direction, and then ask whether a hidden chamber changes that number enough to notice.

Key result 2.1: Predicted muon count along a direction

For a viewing direction with opacity ϱ , the recipe your code follows is: work out the minimum energy $E_{\min}$ a muon needs to survive that opacity (Key result 1.1); add up how many cosmic muons per second, per square metre, arrive from that direction *with at least that much energy*; then multiply by your detector's area and how long it watches. In symbols, for a detector of area A over an exposure time T ,

$$N(\varrho, \theta) \approx AT \times [\text{rate of muons from direction } \theta \text{ with energy } \geq E_{\min}(\varrho)].$$

The bracketed rate comes from the full energy-and-angle spectrum of cosmic muons — a standard, if slightly intimidating-looking, formula that Appendix A gives you in full, and that your code skeleton already implements. You do not need to read that formula, let alone memorise it, to understand what matters here: *raise $E_{\min}$ even a little, and N falls fast*, because the cosmic-ray spectrum drops off steeply with energy — there are far fewer 50 GeV muons raining down than 5 GeV ones.

Watch out It is tempting to use the simple flux law from Key result 1.2 ($I_v \approx 70 \text{ m}^{-2}\text{s}^{-1}\text{sr}^{-1}$ for $p > 1 \text{ GeV}/c$) for this step. Do not — that number already has the 1 GeV threshold baked in, and a real target's opacity demands a much higher threshold (recall Example 1.1: 50 GeV for 100 m of limestone). Using the 1 GeV number will predict wildly too many muons. Use the full energy spectrum, integrated above the actual $E_{\min}$ for each direction — that is precisely what the code skeleton's ray-tracing function is built to do for you.

2.3 Is it real, or did you just get lucky?

A hidden chamber raises the muon count from its direction, compared with the solid rock around it. But muon arrivals are random — even solid rock does not deliver exactly the same count every night. The question every good muography paper has to answer, and the question your report has to answer too, is: *how much of an increase is enough to trust?*

Definition 2.2: Counting statistics

Muon arrivals are independent, random events at a steady average rate — the textbook conditions for a *Poisson process*. If the expected count in some exposure time is μ , the actual count you measure will typically land within $\sigma = \sqrt{\mu}$ of that value, just from statistical luck. Twice as many expected muons means $\sqrt{2}$ times the statistical uncertainty, not twice — which is exactly why longer exposures pay off faster than you might guess.

Key result 2.2: Signal significance

If solid rock in some direction would give you an expected count μ_{rock} , and the same direction with a hidden chamber behind it gives you a higher expected count μ_{void} , a simple and standard way to judge whether the difference is trustworthy is

$$S = \frac{\mu_{\text{void}} - \mu_{\text{rock}}}{\sqrt{\mu_{\text{rock}}}}.$$

This is a simplified, educational significance measure, not a complete statistical treatment of a real muography discovery — a published result would also account for systematic uncertainties, look-elsewhere effects across many pixels, and more. Within that limit, physicists commonly treat $S \gtrsim 3$ as worth taking seriously, and $S \gtrsim 5$ as the conventional discovery threshold — and 5σ , specifically, is the bar your project's headline result is measured against (Chapter 6 says more). S grows with exposure time, because $\mu \propto T$ while $\sqrt{\mu} \propto \sqrt{T}$ — so $S \propto \sqrt{T}$. **This is the formula behind the central question of your project:** how many days, weeks, or months of simulated exposure does it take before S for your chamber design reaches 5σ ?

Example 2.1: A toy significance, worked by hand

Suppose one viewing direction through solid rock is expected to deliver $\mu_{\text{rock}} = 400$ muons over some exposure, and the same direction with a hidden chamber behind it delivers $\mu_{\text{void}} = 460$. Then

$$S = \frac{460 - 400}{\sqrt{400}} = \frac{60}{20} = 3.0.$$

A 15% excess, here, is right at the edge of “worth taking seriously.” Double the exposure time and both counts double, but the significance only grows by $\sqrt{2}$: $S \approx 4.2$. These are made-up numbers — your own chamber, at your own exposure times, will

give you different ones. Finding out what those numbers actually are is the point of week 5 and 6.

Think about it Two designs give the same significance $S = 5$ at their respective “done” exposure times, but one took a week and the other took two months. Which result would you trust more, and why might your report want to say something about both the significance and the exposure time it took to get there, not just one of them?

Intuition Absorption muography asks exactly one question of every direction: *how much stuff is in the way?* It needs only one detector, a lot of patience, and a target you can stand near. (Scattering muography, which asks *what kind* of stuff, instead tracks each muon’s trajectory both before and after it crosses the target — generally requiring detector coverage on both sides — and is not part of this project. But if you finish early, it is a genuinely interesting thing to read about.)

2.4 Model assumptions in this project

Every simulation is built on assumptions, and a good researcher can list theirs. This introductory model assumes:

1. Straight-line muon trajectories through the target (no scattering).
2. A density map with known, exactly-specified density values.
3. Continuous energy loss, using the simplified range–energy relation of Key result 1.1.
4. A specified sea-level cosmic-muon energy spectrum (Appendix A).
5. Poisson counting statistics, with no other source of noise.
6. An ideal detector: perfect efficiency, no dead time, no instrumental background, unless you deliberately choose to model one of these.
7. No atmospheric variation during an exposure (see the note above).
8. Independent angular bins — no coupling between neighbouring directions from a detector’s finite angular resolution.

Note These assumptions are what make an eight-week project possible at all, and, in reverse, the list of what would need to be relaxed for a realistic detector–design study — a good closing paragraph for your report’s discussion section, not a weakness to hide. Chapter 1 told you what falls on you for free; this chapter told you why counting it tells you what is hidden, and how to know when you can trust what you counted. Part II turns both ideas into the actual pipeline you will code, step by step.

Before you move on

- I can define opacity and explain why a hidden chamber sends you more muons, not fewer.
- I understand why the simple 1 GeV flux number is the wrong one to use for a thick target.
- I can state the Poisson rule $\sigma = \sqrt{\mu}$ and explain what it means in plain words.
- I can compute a signal significance from two expected counts, and I have done it once by hand.
- I can list at least three of this model's simplifying assumptions without looking them up.
- Glossary: **opacity, Poisson process, signal significance, exposure time.**

The project

Five steps in code, and the arithmetic that tells you when to
believe them



Your project, in five steps

In a nutshell

In a nutshell. Strip away the physics language and your project is a five-step pipeline: build a fake mountain with a secret inside it, send simulated muons through it from many viewing directions, turn what survives into expected counts and then a noisy dataset, try to see the secret in that dataset, and find out how long the expected signal needs to reach the project's 5-sigma benchmark. The student project sheet maps each step to a deliverable.

Key terms. density map · ray tracing · reconstruction · before-and-after pair · parameter scan.

3.1 What your mentor supplies, and what you build

You are not starting from a blank file. Before week 1, you receive the student package with the supplied physics and geometry utilities, starter notebooks, and a one-page project sheet. Your work is to understand, test, and extend that code. In particular, you implement the two student-owned analysis functions in `code/student_analysis.py`: `opacity_map` and `expected_counts`. The numerical baseline parameters live in `code/project_config.py`; use that file as the single source of truth rather than retyping values in notebooks.

Note “Documented and reproducible,” the standard this project is held to, has a concrete meaning: your final submission should include your source code, the parameter values behind your final figures, and either a fixed random seed or a clearly documented procedure for regenerating your simulated dataset. Because Step 3 below adds *random* Poisson noise, running your own code twice without fixing a seed will not reproduce the same before-and-after image — which is a feature while you are debugging (it shows you what is real signal and what is noise) and a problem if you forget to say so in your report. Use a fixed seed while developing, and state plainly in your methods section whether your final results use one fixed seed or an average over several independent runs.

What a notebook page actually looks like

Every notebook mixes exactly three kinds of cell (the student package's README says this too). 00_start_here.ipynb — the first thing you will open in week 1 — uses all three within its first page. Here it is, cell for cell, with the real plot your own code will produce.

notebooks/00_start_here.ipynb

Explanation --- read it

Does everything work?

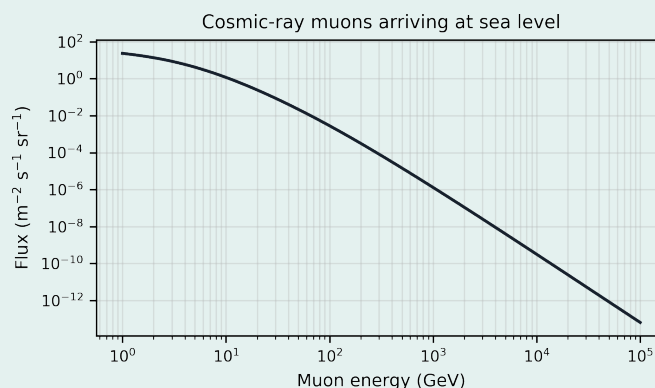
Run the cell below. If you see a plot of a curve falling steeply to the right, you are set up correctly.

The notebooks are designed to be independent. Later notebooks import the shared functions from code/student_analysis.py; they do not depend on variables left in memory by an earlier notebook.

Given code --- run it

```
E = np.logspace(0, 5, 200) # muon energies from 1 to 100,000 GeV
flux = mg.gaisser_flux(E, theta=0.0)
```

```
plt.loglog(E, flux)
plt.xlabel('Muon energy (GeV)')
plt.ylabel('Flux (m$^{-2}$ s$^{-1}$ sr$^{-1}$ GeV$^{-1}$)')
plt.title('Cosmic-ray muons arriving at sea level')
plt.grid(alpha=0.3, which='both')
plt.show()
```



YOUR TURN --- write it

Write a few lines that create 1000 energies between 1 and 10,000 GeV, compute the flux at each, and print how many of them have a flux below $1e-6$.

There is no trick here. It is just to make sure the four ideas above are yours before you need them.

YOUR TURN

Colour carries the same meaning here as everywhere else in this handbook: ink for something to *read*, teal for something already working that you can *trust*, gold for something only you can *finish*.

3.2 Step 1 — Build a target with a secret

You construct a 2D density map: a grid of numbers, each one the density of material at that point, shaped like a mountain or a pyramid, with a low-density patch — a chamber, a cavity, a void — hidden somewhere inside it. You choose the size and depth of the chamber. You are the only one who knows where it is, which is exactly the situation a real muography team is in *before* they look at their data, and exactly the situation your reconstruction algorithm has to work in too.

Think about it If you make the chamber enormous and shallow, the project becomes almost too easy. If you make it tiny and deep, it may be genuinely undetectable in any reasonable exposure time. Somewhere in between is a chamber that is *interesting* — detectable, but only with real statistics behind you. Where do you think that line is, roughly, before you have run a single simulation?

3.3 Step 2 — Trace muons through it

For each viewing direction, you extend a supplied single-ray tracing function to send a muon's path through your density map and add up the density it encountered — the opacity, ϱ , of Definition 2.1. Do this for many directions, covering the whole target, and you get a map of opacity versus direction: high where the target is thick, low over the hidden chamber and near its edges.

Intuition This step is nothing but geometry and addition. The physics does not enter here at all — you are simply asking, for every line of sight, “how much material would a muon travelling along this exact path have to get through?” The physics only enters in the next step, when you decide how many muons actually make it.

Watch out It is easy to trace rays only through the *centre* of the chamber and declare victory. A real detector sees the target from one fixed position, at a whole range of angles — some rays graze the chamber's edge, some miss it by a hair, some pass through only its corner. Make sure your direction grid is fine enough, and wide enough, that the chamber's shape actually shows up in the opacity map rather than being represented by two or three lucky rays.

3.4 Step 3 — Turn opacity into expected counts, then add noise

For each direction, convert the opacity into a minimum survival energy (Key result 1.1), then into a predicted muon count using the full energy spectrum (Key result 2.1). That predicted count is the *expectation*: the average you would obtain if you repeated the same exposure many times. A simulated observation is one Poisson random draw from that expectation (Definition 2.2). Weeks 5 and 6 separate these ideas deliberately: the clean ratio uses expected counts; Poisson noise is added afterwards to demonstrate statistical fluctuations.

Watch out Do this step twice: once for your target with the chamber in it, and once for a twin target with the chamber filled back in solid — your target's own without_void(). You will need both maps in Step 4, and it is easy to forget the second one until you are already stuck.

3.5 Step 4 — Reconstruct, and ask ``is it there?''

Definition 3.1: 2D muon transmission radiograph

What you are building is not a full tomographic reconstruction (which would infer a complete 3D density model from many overlapping views). It is simpler and older: one viewing direction becomes one pixel or bin, directly, the way Alvarez's original pyramid image and the ScanPyramids figures work. That simplicity is a feature for an introductory project — it is honest to call it a *muon radiograph*, and worth saying so in your report. One caveat worth carrying into your discussion section: in the simulation, each angular bin is a clean, independent pixel. A real detector has finite angular resolution and an acceptance that blurs and couples neighbouring directions together — a real muon radiograph is never quite as crisp, pixel for pixel, as this toy model's.

The first, most obvious thing to try is to plot the raw counts from step 3 directly, one pixel per direction, and look at it.

Watch out This does not work, and it is meant not to. A raw-counts image is dominated by one large, boring, entirely predictable effect: your target is thicker in some directions than others, simply because of its shape. In a typical pyramid geometry the shape changes the count by a factor of ten between the thinnest and thickest directions; the chamber changes it by tens of per cent. Plot raw counts and you will see a pyramid, not a chamber. If your first image looks like nothing but a gradient, that is not a bug — it is the finding, and it is exactly why the next step exists.

Key result 3.1: Dividing out the shape of the target

The fix is the standard one for this kind of problem in imaging generally: *predict the boring part, and divide it out*. Take the ratio, pixel by pixel,

$$\text{ratio} = \frac{\text{counts through the target with the chamber}}{\text{counts through the same target with the chamber filled solid}},$$

using the twin target from the pitfall in Step 3. Away from the chamber this ratio is flat at 1.0 — the target’s shape has cancelled out of both the numerator and the denominator. Over the chamber, it rises above 1.0 by however much the missing rock let extra muons through. This ratio map, not the raw-counts map, is the “after” half of your before-and-after pair (Definition 3.2): it is the image that actually shows the chamber.

Definition 3.2: The before-and-after pair

Before: the true density map you built in step 1, with the chamber visible because you drew it there yourself. **After:** the ratio map of Key result 3.1 — or, once noise enters the picture below, the significance map built from it — with no knowledge of where the chamber is beyond what the muons themselves reveal. The entire point of the project is the gap between these two images, and how that gap survives, or does not, once real statistics are involved.

Definition 3.3: Expected and observed significance

Expected significance compares the expected count with a chamber to the expected count with the chamber filled in:

$$S_{\text{exp}} = \frac{\mu_{\text{void}} - \mu_{\text{rock}}}{\sqrt{\mu_{\text{rock}}}}.$$

It is deterministic for fixed model parameters and defines the smooth exposure curve used for the headline result. **Observed significance** uses one Poisson-noisy count map instead, so it varies with the random seed. The two agree on average, but an individual noisy observation is not expected to lie exactly on the smooth curve.

For the official exposure study, first use the *expected* counts and compute the expected significance for a fixed signal direction. This gives a smooth, deterministic curve. Then compare that curve with one or more Poisson-noisy observations to see how an actual simulated measurement fluctuates around it. Do not choose a new signal pixel from each noisy map: that would introduce a selection bias.

Example 3.1: Reading a signal-significance curve

A typical expected-significance curve rises as $\sqrt{T}$. The exposure time where the *expected* curve first crosses $S = 5$ is the headline number for this project: the estimated

exposure time of the baseline simulated setup under the stated model assumptions. Mark $S = 3$ too as an earlier waypoint. A noisy observation will fluctuate around the smooth curve, so its 5-sigma crossing is not the official exposure-time definition.

3.6 Step 5 — Change one thing at a time

Once the pipeline works for one chamber, in one place, at one exposure time, the interesting science starts: change one thing about the setup, one at a time, and watch how the required exposure time moves.

Key result 3.2: Four studies, pick two

Your code skeleton frames this step as four named studies. You choose **two** — two done carefully beats four done in a hurry, and your report has room for two.

- **Study A — chamber size.** Vary the chamber's radius at fixed depth and exposure. Volume goes as r^3 ; the path length a muon's line of sight loses by passing through the chamber goes as r . Which one governs how significance scales with size? Write down a prediction before you run it.
- **Study B — chamber depth.** Move the chamber up and down at fixed size and exposure. Two effects fight each other: less rock above a shallower chamber, but a longer, more slanted path from the detector below. Which wins?
- **Study C — detector placement.** Keep the chamber fixed and move the detector. This is Definition 3.4 put into practice, and it is the question a real muography team must answer before committing months of fieldwork to one spot.
- **Study D — angular bin size.** Coarser angular bins collect more muons per bin but redistribute the chamber signal over fewer, larger pixels. This is a simplified binning study, not a full model of a detector point-spread function; state that limitation in your report.

Whichever two you pick, the method is the same for all four: change that one thing, hold everything else fixed, and plot the effect on the required exposure time or the significance curve.

Definition 3.4: Detector placement

In this project, *placement* means the detector's position relative to the target: how far away it sits, and at what height or offset, which between them set which lines of sight through the target it actually covers. Unless you have a specific reason to do otherwise, orient the detector so its field of view covers the target and the chamber inside it — a placement that cannot see the chamber at all cannot detect it, no matter how long you wait.

Example 3.2: The parameters you need to pin down

Before any Step 5 study means anything, write down — and keep fixed unless it is the one thing you are deliberately varying — every quantity below. Your mentor’s code skeleton will supply working defaults for most of these; your job is to know what they are, not to guess.

PARAMETER	WHAT IT SETS	TYPICALLY
Detector area	how many muons you collect per second (bigger area, shorter exposure)	fixed
Exposure time	the x -axis of the expected-significance curve	varied (Step 4)
Chamber size	Study A	varied only in Study A
Chamber depth	Study B	varied only in Study B
Detector placement	Study C	varied only in Study C
Angular binning	Study D	varied only in Study D
Rock/target density	sets the opacity scale outside the chamber	usually fixed
Random seed	whether your noisy dataset is reproducible	fixed while debugging

Note Exposure time scales inversely with detector area — double the area and you halve the time needed for the same statistics — so “how many days” is never a question you can answer without first knowing what detector area the question assumes.

Watch out Vary two things at once — chamber size *and* depth, say — and you will not know which change caused which effect in your plot. Every one of the four studies above changes exactly one parameter; do not be tempted to combine two into one faster study. This is the single most common mistake in a first research project, in any field, and it is entirely avoidable.

3.7 Where the real literature comes in

A bit of history Your reconstructed image, your exposure study, and your discussion of detectability are not just numbers for their own sake — they address questions closely related to what the 2017 ScanPyramids team actually faced with the real Khufu void (Morishima *et al.*, *Nature* **552**, 386): is there a void here, and how do you know? Your simulation is a simplified model, not a reproduction of their experimental analysis, so keep the comparison at the level of physics and scale — how your toy numbers behave in the same *kind* of way theirs did — rather than treating your figures as standing in for their result. That honest framing, not “I built a simulation,” is what turns a coding exercise into a short scientific report.

Note Five steps, and every deliverable in `docs/project_sheet.md` is one of them written up properly: the working simulation is steps 1–3, the before-and-after pair and significance study are step 4, the comparison figures are step 5, and the report and slides are all five, told as a story. Chapter 4 covers the tools and plotting habits that make each step easy to trust.

Before you move on

- I can describe, in five sentences, exactly what my code will do in each step.
- I understand why step 2 (ray tracing) is pure geometry, and step 3 (counts) is where the physics enters.
- I know what a signal-significance curve is and what its x-axis and y-axis mean.
- I have a plan to change one parameter at a time in step 5, not several together.

Python, plots, and the arithmetic of trust

In a nutshell

In a nutshell. You need three things to do this project well: Python well enough to trust your own arithmetic, a habit of making plots that a stranger could read without you standing next to them, and a running answer to the question every real scientist asks constantly — *how do I know this isn't just noise?* None of these require you to be an expert programmer. They require you to be careful.

Key terms. NumPy · Matplotlib · vectorised code · labelled axes · error bars · reproducibility.

4.1 The tools, and no more than you need

This project uses Python with two libraries: `numpy` for arrays and arithmetic, and `matplotlib` for plots. You do not need C++, you do not need Geant4 or ROOT (the professional-grade tools this group uses for its real detector work), and you do not need to touch a command line beyond what it takes to start a Jupyter notebook. Installation instructions are in Appendix B; do that *before* week 1 starts, not during it.

Example 4.1: Your first real plot: flux versus zenith angle

Task 1 of week 1 asks you to reproduce the flux law of Key result 1.2 as a plot. In plain NumPy and Matplotlib, that whole exercise is genuinely this short:

```

import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi/2, 200)    # zenith angle, 0 to 90 deg
I_v = 70                                # m-2 s-1 sr-1
I = I_v * np.cos(theta)**2

plt.plot(np.degrees(theta), I)
plt.xlabel("Zenith angle (degrees)")
plt.ylabel(r"Flux  $I(\theta)$  (m-2 s-1 sr-1)")
plt.title("Cosmic muon flux vs. zenith angle")
plt.tight_layout()
plt.savefig("flux_vs_angle.png", dpi=300, bbox_inches="tight")
plt.show()

```

A dozen lines. If you can write, run, and understand every line of this, you are ready for week 1. If you cannot yet, that is exactly what week 1's intro tutorials are for. The last three lines are worth making a habit of for every figure you ever save: `tight_layout()` stops labels being clipped, `dpi=300` with `bbox_inches="tight"` is what a report figure actually needs (150 dpi is fine for a quick look on screen, not for something you hand in), and `show()` lets you check a plot before you trust it.

Intuition NumPy arrays let you write one line of arithmetic — `I = I_v * np.cos(theta)**2` — and have it apply to two hundred angles at once, instead of writing a loop. This is not just shorter; it is faster, and it is much harder to get subtly wrong. Once your density map and your direction grid are both NumPy arrays, tracing many rays through the target becomes the same kind of one-line-does-a-lot arithmetic, not two hundred separate calculations you have to keep straight in your head.

4.2 Making a plot someone else can trust

Definition 4.1: A plot that earns its place in a report

Every figure you hand in should have: axes that are labelled, with units; a title or caption that says what the figure shows, in words, not just a filename; and, wherever the number has a statistical uncertainty, an error bar showing it. A plot without these is a picture. A plot with them is evidence.

Watch out The most common way a first research plot fails is silence about the y-axis: numbers with no units, no scale, and no sense of whether the reader is looking at 10 or 10,000. Before you hand in any figure, ask yourself whether someone who has never seen your code could look at it and say, correctly, what it shows.

4.3 Keeping your work honest and recoverable

Note Keep a short, dated log of what you tried and what happened — even three lines a day. “Tuesday: chamber at depth 40, S never crossed 3 even at 90 days — tried moving detector directly above it instead of off to the side” is worth more, a month later, than a folder of unlabelled files called `test2_final_v3.py`. Save your notebook (or your script and its output plots) somewhere your mentor can see it between meetings — a shared folder, an email, or a small git repository if you already know one. The tool matters far less than the habit.

4.4 Using AI tools honestly

The project sheet lists “AI Tools” among the useful skills, and it is entirely reasonable to ask an AI assistant to explain a NumPy error, sanity-check a formula, or suggest why a plot looks wrong. What it cannot do for you is understand your own chamber, your own significance curve, or your own answer to “how many days.” If you cannot explain, in your own words and without looking anything up, why your reconstructed image looks the way it does, you are not ready to write that part of the report yet — no matter who or what wrote the code.

Definition 4.2: AI tools: allowed, but accountable

You may use AI to: explain a Python concept, debug an error message, suggest ways to visualise your data, explain unfamiliar terminology, or improve your English writing.

You must not: submit AI-generated results you have not checked yourself, treat AI-generated physics as automatically authoritative, let it invent a reference or a numerical result you have not verified, or submit code you cannot explain line by line.

If an AI tool materially contributed to a piece of code or an argument in your analysis, keep a short note of what it was used for — the same habit as citing any other source you leaned on.

Think about it If an AI assistant wrote a line of your simulation for you, and it turns out to be subtly wrong, whose mistake is it in the report? There is only one honest answer, and it is worth deciding on it now, before week 6.

4.5 How to ask for help

Definition 4.3: The one-hour rule

Stuck for an hour? Message your mentor. Stuck for a day in silence? That is the one real mistake available to you in a remote project like this one — nobody can see you frowning at your screen.

A good question contains four things: what you were trying to do, what you did, what you expected, and what actually happened — including the exact error message, copied, not paraphrased or summarised. A question with those four things is usually answered in minutes. “It doesn’t work” takes much longer, for everyone.

Before you move on

- Python, Jupyter, NumPy and Matplotlib are installed and working on my own computer.
- I can write the flux-vs-angle plot from Example 4.1 without copying it.
- I know what makes a plot “earn its place” in a report, and I check every figure against it.
- I am keeping a short dated log of what I try, and I know where my mentor can see my work.

Your eight weeks

The schedule, what you hand in, and getting ready before
week one



The plan

In a nutshell

In a nutshell. Eight weeks, eight hours a week, one project, three of you working it independently in parallel with a weekly meeting with your mentor. The five steps of Chapter 3 map almost exactly onto eight weeks of work — this chapter just puts dates and a “what done looks like” next to each one. Exact dates are confirmed by your mentor; the shape of the plan is not.

Key terms. milestone · weekly meeting · scope · the code skeleton.

5.1 Eight weeks, week by week

WEEK	FOCUS	WHAT “DONE” LOOKS LIKE
1	Orientation; Python tutorials; what a muon is	NumPy/Matplotlib tutorials complete. You have reproduced the flux-vs-zenith-angle plot (Example 4.1) yourself, and you can explain why muons penetrate rock.
2	The range–energy relation; a 1D survival model	You can calculate, by hand and in code, how many muons of a given energy survive a given thickness of rock. Verified against Example 1.1 for at least one test case.
3	Build your virtual target	A 2D density map exists: a mountain or pyramid shape, with a hidden low-density chamber of a size and depth you have chosen. The supplied single-ray function traces one path through it correctly.
4	Ray-trace the whole target	The single-ray function is extended to many directions covering the target. You have an opacity map: high where the target is thick, low over the chamber.
5	Counts and a first image	Opacity is converted to <i>expected</i> muon counts (Key result 2.1) for the target with and without its chamber. The clean ratio (Key result 3.1) gives your first reconstructed image — your before-and-after pair (Definition 3.2). Poisson noise is introduced in week 6.

WEEK	FOCUS	WHAT "DONE" LOOKS LIKE
6	Noise, significance and exposure time	Poisson noise is added to the expected counts, observed significance maps are produced, and the expected-significance curve is built at several exposure times (a day, a week, a month, ...). You determine the first 5σ crossing of the expected curve.
7	Two parameter studies	Two of the four named studies (Key result 3.2: chamber size, chamber depth, detector placement, angular bin size) run to completion, one parameter varied at a time within each (never two together — see the pitfall in Chapter 3). Findings discussed with your mentor and compared against the 2017 Khufu result. A first draft of the report, with figures in place, is a good target to leave this week with.
8	Report, slides, demo	Figures finalised. Report written. Slides built and rehearsed once, out loud, before demo day.

Note Three of you are on this project, and each completes the full scientific pipeline independently. You are not dividing the project into separate jobs: all three start from the same supplied starter code, and each owns their own understanding, tests, chamber design, and results. In Week 7 you may informally explore different parameters first (for example, size, depth, or detector placement) and compare notes, but each student completes their own chosen scan. Discuss the division with your mentor in Week 6.

Watch out **Week 4 is the gate.** If your ray tracing is not producing a sensible opacity map — high over thick rock, low over the chamber — by the end of week 4, everything downstream (counts, noise, images, significance) will inherit the same bug and be much harder to debug later, buried under three more layers of code. Get a colleague, your mentor, or a fresh pair of eyes on it in week 4, while the bug is still shallow.

Think about it The plan above assumes eight hours a week is spent roughly evenly. In practice, some weeks (debugging the ray tracer) eat more hours than others (writing the report, once the figures already exist). Is that a problem, or is that just what research time actually looks like?

Before you move on

- I know what week I am supposed to be finishing, and what “done” looks like for it.
- I understand why week 4 is the gate, and what I will do if my opacity map does not make sense by then.
- I have a specific chamber size and depth chosen, and I know why I picked them.

What you hand in

In a nutshell

In a nutshell. Seven things, six required and one optional, each with a standard attached. None of them is a surprise — every one of them is a direct write-up of a step from Chapter 3. The report and the slides are not extra work on top of the simulation; they are the simulation, explained to someone who has not seen your code.

Key terms. reproducible · scientific report · demo day · stretch outcome.

6.1 Everything that is due

DELIVERABLE	BY	THE STANDARD
Working simulation	Week 6	Documented and reproducible: traces muons through a density map you define, and returns expected counts and Poisson-noisy observations for each direction. Somebody else could re-run the notebooks from fresh kernels using the shared student module and get equivalent results for a fixed seed.
Before-and-after pair	Week 5	The true target with its hidden chamber, next to the clean ratio map recovered from the expected counts (Definition 3.2). The gap between the two is your evidence.
Significance curve	Week 6	How detectability improves with exposure time, plus the project's headline number: the estimated exposure time, <i>under the simulation model</i> , at which the <i>expected chamber signal</i> first reaches 5σ . The smooth expected-significance curve defines this number; noisy observations fluctuate around it. Discuss how a real detector's acceptance, efficiency, resolution, and atmospheric variation would change that estimate.
Comparison figures	Week 7	Two of the four named studies from Key result 3.2 (chamber size, chamber depth, detector placement, angular bin size), your choice, each varying one parameter and showing its effect on the required exposure time.

DELIVERABLE	BY	THE STANDARD
Written report	Week 8	Scientific format, in $\LaTeX$, using the supplied docs/report_template.tex: abstract, introduction, method, validation , results, discussion, conclusion, and code availability. Situated against published muography work, including the 2017 Khufu measurement.
Slide presentation	Week 8	Suitable for demo day, including a live or recorded run of your simulation generating an image from the starter code and your analysis functions.
Stretch (optional)	If time allows	Apply the simulation to a real candidate target — a Malaysian limestone hill or cave system is the natural choice — with an estimated required exposure time.

6.2 How to write the report

Not an essay — a record, detailed enough that someone who was not here could repeat what you did and get what you got. Use the supplied docs/report_template.tex: it already has the right section headings and comments telling you what belongs in each one, so this section explains the *why* behind that structure rather than repeating it heading by heading.

- **Abstract.** What you simulated, what you found, what it means. Four sentences. Write it *last*, and rewrite it at least twice.
- **Introduction.** Why muography matters and what was already known — this is essentially Chapter 2 of this handbook, in your own words, with the 2017 Khufu result as your anchor example.
- **Method.** Your target, your ray-tracing approach, the exact energy-loss and flux formulae you used (Appendix A gives you the precise forms to cite), and how you turned opacity into counts and counts into an image — in enough detail that it could be rebuilt from your description alone.
- **Validation.** The section a description does not have and a report does: what did you check, and against what? A hand calculation for one ray, a run on `without_void()` that came back flat, a step-size halved to confirm the answer did not move — say what you checked and to what precision it agreed. This is the single biggest difference between “I wrote some code” and a report your mentor can trust without re-deriving everything themselves.
- **Results.** The figures: before-and-after pair, significance curve, your two comparison studies. Each figure makes exactly one point, and each has labelled axes with units (Definition 4.1).
- **Discussion.** What your headline number means, what would make it more realistic (a real detector’s acceptance and efficiency, atmospheric effects, a non-uniform back-

ground target, the look-elsewhere effect across every pixel in your image), and — honestly — what you would do next with another month. Saying what you did not have time for is the most professional paragraph in the report, not a weakness.

- **Conclusion.** Three sentences. No new information.
- **Code availability.** Point to your repository, and say which notebook produced which figure.

Watch out The report is written in week 8, which means the figures must exist by week 7, which means the pipeline must work by week 6. There is no week in this plan set aside for “catching up.” If you start writing on the last day with no finished figures, you will hand in something you are not proud of — and everyone, including you, will know it.

6.3 How to give the talk

- One idea per slide. If a slide needs a full paragraph of speech to explain it, it is really two slides.
- Every plot: point at the axes first, then say what the viewer should see in it. Never assume the audience can read a dense plot as fast as you can.
- Include a live or recorded run of the simulation. Watching a fake mountain turn into a fake muon image, on screen, is more convincing than any bullet point describing the same thing.
- Rehearse out loud, at least once, in front of someone — your mentor, a parent, a rubber duck. The talk you have only rehearsed silently in your head is always longer than the one you actually give.

Think about it What is the one question about your project you most hope nobody asks? Prepare the answer to that one first. It is usually the question that gets asked.

Before you move on

- I know all six required deliverables and roughly when each is due.
- My report will have labelled, unit-bearing axes on every figure, and I have checked this on a draft already.
- I have planned week 7's figures and week 8's report as separate tasks, not one task done at the last minute.

Before week one

In a nutshell

In a nutshell. Get Python, Jupyter, NumPy and Matplotlib installed and working *before* week 1 starts, so that week 1 gives you a week of physics instead of a week of package managers. Then: how the weekly rhythm works, and – honestly – what is going to go wrong, because a chunk of your eight weeks will go to problems that never appear anywhere in your final report.

Key terms. week zero · weekly meeting · logbook · scope.

7.1 Week zero: do this before week one

ITEM	DETAIL
Install	Python 3.11+ with numpy 2.x, matplotlib 3.7+, imageio, and Jupyter. Commands are in Appendix B – it takes about half an hour, and it is much better spent at home than on the first call with your mentor.
Account	An Overleaf account, for the $\text{\LaTeX}$ report in week 8. Free, and worth being comfortable with before you are under deadline pressure.
Read	Chapters 1 and 2 of this handbook, before your first meeting.
Confirm	Your mentor's preferred meeting time, and which communication channel (email, chat, video call) you will use between meetings.

Watch out Do the installation in week zero, not in the first ten minutes of your first meeting. If something defeats you, bring the mentor the exact *error message*, not the conclusion that “it doesn't work” – see the one-hour rule (Definition 4.3).

7.2 The rhythm that keeps eight weeks from evaporating

- **Weekly meeting.** One update: what you did, what you are doing next, and what is blocking you. Never skip the third part — it is the most useful one for your mentor.
- **Every session — a short log.** Date, what you changed, what you saw, what you concluded. Three lines is enough; see the note in Chapter 4.
- **Between meetings — the one-hour rule.** Stuck for an hour, ask. Stuck for a day in silence, that is the mistake, not the being stuck.

7.3 What is going to go wrong

Watch out Your ray tracer will miss the chamber entirely on your first attempt, silently, because a coordinate was flipped. Matplotlib will produce a plot with no visible line because you forgot to call `plt.show()` or save the figure. A NumPy broadcasting error will produce a cryptic message about array shapes that means something completely different from what it sounds like. Your significance curve will look wrong for an hour before you realise you were computing $\sqrt{\mu_{\text{void}}}$ instead of $\sqrt{\mu_{\text{rock}}}$ in the denominator.

A real share of your eight weeks will go to problems exactly like these — problems that will never appear in your final report, because by the time you write it, they are fixed. This is not a sign that the project is going badly. It is the job. The difference between a beginner and a researcher is not that the researcher's code works on the first try — it is that the researcher expects it not to, changes one thing at a time, writes down what they changed, and keeps going.

Note Three rules that will carry you through all eight weeks. **One hour** — ask before a session is lost to silence. **One variable** — change one thing at a time, and write down what you changed; two changes and a working plot together teach you nothing about which one mattered. **One standard** — a null result, honestly reached, is a result; a positive result you cannot reproduce the next day is not.

Eight weeks is short. It is long enough to build one working simulation from scratch, defend a number you calculated yourself, and be asked hard questions about it by people who genuinely want to know the answer. Bring the curiosity that got you into this project, and the willingness to be wrong out loud in front of your mentor. See you in week one.

Before you move on

- Python, Jupyter, NumPy and Matplotlib are installed and working on my own machine before week 1.
- I have an Overleaf account for the week 8 report.
- I have read Chapters 1 and 2 before my first meeting.

- I understand the one-hour rule, and I have decided, in advance, to actually follow it.
- I have accepted, in advance, that a real share of this project will not work on the first try.
Good — that means it is a real project.

Formulae you'll actually use

In a nutshell

Why this is at the back. The main text moves; formulae do not. Everything quoted in Parts I and II is collected here, with the assumptions it rests on, so that when your report needs a citation for a number, this is where you point.

A.1 Energy loss and range

For a particle moving through matter in the region where its ionisation energy loss is near its minimum — cosmic muons at GeV energies are close to this regime — the energy loss rate is well described by

$$\left\langle -\frac{dE}{dx} \right\rangle \approx a + bE, \quad a \approx 2 \times 10^{-3} \text{ GeV g}^{-1}\text{cm}^2, \quad b \approx 4 \times 10^{-6} \text{ g}^{-1}\text{cm}^2,$$

where the constant term a is ionisation (it dominates at low energy) and the term bE is radiative loss — bremsstrahlung, pair production, photonuclear interactions — which only matters at high energy. The ratio $a/b \approx 500 \text{ GeV}$ is the *critical energy*: the point at which the two mechanisms contribute equally, consistent with the few-hundred-GeV figure quoted for standard rock in the literature.

The exact range–energy relation

Solving $dE/dX = -(a + bE)$ gives the minimum energy a muon needs to survive an opacity ϱ exactly, not just approximately:

$$E_{\min}(\varrho) = \frac{a}{b} (e^{b\varrho} - 1).$$

This is the formula your code skeleton actually implements (`minimum_energy` in `muography.py`), and the one your report's methods section should quote. For small opacity ($b\varrho \ll 1$) it reduces to the simple linear rule used in Chapter 1 for intuition, $E_{\min} \approx a\varrho$ (Key result 1.1) — the two agree to within a few percent for a typical muon's own range, and diverge more as the opacity grows into the hundreds-of-GeV regime. Quick sanity check, useful for testing

your own code: 1 m of standard rock ($\varrho \approx 265 \text{ g cm}^{-2}$) needs about 0.53 GeV; 100 m needs about 56 GeV; 500 m needs about 350 GeV.

Watch out Chapter 1's worked examples use the simple linear rule $E_{\min} \approx \varrho \times a$ because it is easier to do by hand and the difference is small at those opacities. Your code does not take that shortcut — it solves the full expression above. If you check your code's output against a hand calculation using the linear rule, expect agreement to a few percent, not to the last decimal place; that gap is the radiative term, not a bug.

A.2 Opacity

$$\varrho = \int \rho d\ell \quad [\text{g cm}^{-2}],$$

the density integrated along a straight-line path through the target — exactly what your ray-tracing code accumulates for each direction.

A.3 Cosmic muon flux

The simple rule, for muons above 1 GeV/ c

$$I(\theta) \approx I_v \cos^2 \theta, \quad I_v \approx 70 \text{ m}^{-2} \text{s}^{-1} \text{sr}^{-1},$$

the origin of “about one muon per square centimetre per minute.” This is the formula for week 1's flux-versus-angle plot. It is *not* the formula to use for predicting how many muons survive a real target — see the pitfall in Chapter 2.

The full spectrum, for converting opacity into a predicted count

The starting point is the classic analytic approximation to the differential cosmic muon spectrum at sea level, due to Gaisser (1990):

$$\frac{d^2 I}{dE d\Omega}(E, \theta) \approx 0.14 \left(\frac{E}{\text{GeV}} \right)^{-2.7} \left[\frac{1}{1 + \frac{1.1E \cos \theta}{115 \text{ GeV}}} + \frac{0.054}{1 + \frac{1.1E \cos \theta}{850 \text{ GeV}}} \right] \text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1} \text{GeV}^{-1},$$

with the two terms in brackets coming from muons produced by decaying pions and kaons respectively. As written, this is a high-energy approximation — properly valid only for $E \gtrsim 100 \text{ GeV} / \cos \theta$ and $\theta \lesssim 70^\circ$ — and it badly *over*-predicts the flux below about 10 GeV, which matters because your target's thinnest corners can have an $E_{\min}$ well under that.

Key result A.1: The spectrum your code actually uses

Your code skeleton does not use the raw formula above. It applies the low-energy correction of Guan *et al.* (2015) (see Appendix C), which replaces E in the expression

above with an effective energy

$$E_{\text{eff}} = E \left(1 + \frac{3.64 \text{ GeV}}{E (\cos \theta)^{1.29}} \right)$$

before evaluating it. This single change keeps the formula accurate down to low energy and moderate zenith angle, at the cost of no longer being the “textbook” Gaisser formula quoted in older references. Integrating the corrected spectrum from $E_{\text{min}}(\vartheta)$ to infinity, and multiplying by detector area and exposure time, gives the predicted count of Key result 2.1. This is also why the number you get from `mg.integrated_flux(1, 0.0)` in notebook 01 (around $60 \text{ m}^{-2}\text{s}^{-1}\text{sr}^{-1}$) is close to, but not exactly, the simple $I_v \approx 70$ of the rule of thumb above: they are two different, both legitimate, approximations to the same real quantity.

Watch out Do not go looking for a bug if your code’s flux numbers do not match a hand calculation done with the raw Gaisser formula — they are not supposed to. This group’s own detector-design work uses the same family of corrected parametrisations (Guan-modified Gaisser, cross-checked against MCEq and full Monte Carlo), through dedicated flux-modelling software; your project uses a lighter version of exactly that approach, not a simplification of it that a “more realistic” calculation would need to replace.

A.4 Counting statistics and significance

$$\sigma = \sqrt{\mu}, \quad S_{\text{exp}} = \frac{\mu_{\text{void}} - \mu_{\text{rock}}}{\sqrt{\mu_{\text{rock}}}}, \quad S_{\text{exp}} \propto \sqrt{T}.$$

As Key result 2.2 says: this is a simplified educational significance measure, not a full statistical treatment. For this project, S_{exp} is the *expected significance* and defines the smooth exposure curve used for the headline result. The 3σ and 5σ levels are project benchmarks. Your code’s `significance()` instead uses a noisy observed count, while `expected_significance()` compares two expectation maps. The two quantities agree on average, but an individual noisy observation will jitter around the expected curve.

Installing Python

B.1 Install

```
# --- Option 1: conda / miniconda (Windows, macOS, Linux) ---
conda create -n muography python=3.11 "numpy>=2,<3" "matplotlib>=3.7" jupyter imageio
conda activate muography

# --- Option 2: plain pip, if you already have Python 3.11+ ---
python3 -m venv muography

# activate it -- the command depends on your shell:
source muography/bin/activate          # macOS / Linux
muography\Scripts\activate             # Windows, Command Prompt
.\muography\Scripts\Activate.ps1      # Windows, PowerShell

pip install "numpy>=2,<3" "matplotlib>=3.7" jupyter imageio

# --- Verify ---
python -c "import numpy, matplotlib; print('all good')"
jupyter notebook                       # should open in your browser
```

Watch out Do this in week zero, at home, with time to spare. If it defeats you, message your mentor with the exact error message before your first meeting — not during it.

Reading

Note The full citations below are deliberately complete — author list, title, journal, volume, pages, year, DOI — not just “Morishima et al.” This is also, quietly, what a properly formatted scientific reference looks like; your own report’s bibliography should follow the same pattern.

C.1 The physics

- Morishima, K., Kuno, M., Nishio, A. *et al.* *Discovery of a big void in Khufu’s Pyramid by observation of cosmic-ray muons.* *Nature* **552**, 386–390 (2017). doi:10.1038/nature24647. The paper your project is, in miniature, recreating the method of.
- Alvarez, L. W. *et al.* *Search for Hidden Chambers in the Pyramids.* *Science* **167**(3919), 832–839 (1970). doi:10.1126/science.167.3919.832. The original pyramid muography paper, fifty years before Morishima *et al.*
- Bonomi, G., Checchia, P., D’Errico, M., Pagano, D., Saracino, G. *Applications of cosmic-ray muons.* *Progress in Particle and Nuclear Physics* **112**, 103768 (2020). doi:10.1016/j.pnpnp.2020.103768 — a readable review of muography, covering pyramids, volcanoes, and much else besides.
- Navas, S. *et al.* (Particle Data Group). *Review of Particle Physics.* *Phys. Rev. D* **110**, 030001 (2024), sections *Passage of particles through matter* and *Cosmic rays*. Free online, authoritative, updated every two years, and worth bookmarking.
- Gaisser, T. K. *Cosmic Rays and Particle Physics.* Cambridge University Press (1990) — the original source of the flux formula in Appendix A.
- Guan, M., Chu, M.-C., Cao, J., Luk, K. B., Yang, C.-g. *A parametrization of the cosmic-ray muon flux at sea-level.* arXiv:1509.06176 (2015) — the low-energy correction to Gaisser’s formula that your code skeleton actually implements (Key result A.1).
- Tang, A., Horton-Smith, G., Kudryavtsev, V. A., Tonazzo, A. *Muon simulations for Super-Kamiokande, KamLAND, and CHOOZ.* *Phys. Rev. D* **74**, 053007 (2006). doi:10.1103/PhysRevD.74.053007 — a different, independently derived modified Gaisser

parametrisation with the same aim; mentioned so that if you read other muography papers you recognise the family, not because your code uses this particular one.

C.2 The tools

- NumPy quickstart: <https://numpy.org/doc/stable/user/quickstart.html>
- Matplotlib tutorials: <https://matplotlib.org/stable/tutorials/index.html>
- Overleaf (for the week-8 L^AT_EX report): <https://www.overleaf.com>

C.3 If you reach the stretch goal

Malaysia has no shortage of limestone: Batu Caves and the Gunung Mulu cave system are the best-known examples, but any hill with known cave passages or a suspected void is a fair candidate. You will not have real detector data for it — the stretch goal asks only for a plausible density model and an honest estimate of the exposure time a real survey would need, using exactly the pipeline you already built.



Glossary

TERM	MEANING
Muon	A heavier, unstable cousin of the electron – same charge, 207 times the mass, minimum-ionising.
Minimum ionising particle (MIP)	A charged particle whose rate of energy loss is near the lowest it can be; cosmic muons are the textbook example.
Range	The mass thickness of material a particle of a given energy can cross before stopping.
Zenith angle (θ)	The angle measured from straight overhead.
Flux	The rate of particles arriving per unit area, per unit time, per unit solid angle.
Opacity (ϱ)	Density integrated along a path through a target; the “how much stuff is in the way” number.
Absorption muography	Imaging by counting how many muons survive a path through a target; contrast with scattering muography, which is not part of this project.
Ray tracing	Computing the opacity along a straight-line path through a density map.
Poisson process	A random process of independent events at a steady average rate; muon arrivals are a textbook example.
Expected significance (S_{exp})	The deterministic signal-to-statistical-noise measure used for the project headline; $S_{\text{exp}} = (\mu_{\text{void}} - \mu_{\text{rock}}) / \sqrt{\mu_{\text{rock}}}$.
Observed significance (S_{obs})	The same educational statistic evaluated with one noisy Poisson count map in place of the expected chamber count.

TERM	MEANING
Exposure time	How long a (real or simulated) detector collects data.
Ratio map	Expected or observed counts with the chamber divided by counts for the same target with the chamber filled in; the chamber appears as an excess above 1.
Angular bin size	The width of the angular pixels in this simulation; it is not the same thing as a real detector's physical angular resolution.
Reconstruction	Turning measured (or simulated) muon counts back into an image of the target.
Before-and-after pair	The true density map next to the image reconstructed from simulated data alone.
Parameter scan	Varying one quantity at a time (chamber size, depth, or detector placement) and recording how the result changes.